

Learning Intention: I can explain how scientists use data from observations to develop explanations and make predictions.

Success Criteria — I can:

- Explain what data is and how it is collected.
- Describe how scientists use data to explain what they observe.
- Explain how data can be used to make a prediction.

Reading Strategy

Before reading, look at the heading and any bold words — they tell you what the text is about. As you read, look for **examples** of people collecting data and **reasons** why that data is useful.

Read — Scientists and Their Data

Scientists don't just guess how the world works — they collect **data**. Data is information gathered from careful observation, measurement, or experiments. Scientists use data to build explanations they can trust.

A weather scientist, called a **meteorologist**, records daily data like temperature, wind speed and rainfall. By looking at patterns in this data over many years, meteorologists can explain why certain weather happens and predict what tomorrow's weather might be like.

A wildlife ranger might count how many turtles hatch on a beach each year. By comparing this data year after year, the ranger can explain whether the turtle population is growing or shrinking, and predict what might happen if nothing changes.

Doctors and scientists studying health collect data too — like how many people recover from a certain illness after trying a new treatment. This data helps them explain whether the treatment works, and predict how well it might work for other patients in future.

In every case, data is the evidence that turns a guess into a scientific explanation.

Vocabulary Helper

Word	Meaning
Data	Information gathered from observation or measurement
Explanation	A statement that makes sense of what has been observed
Prediction	A sensible guess about what will happen, based on data

Parent Tip

Keep a simple weather diary together for a week — record the temperature or whether it rained each day. At the end of the week, look at the data together and talk about any patterns you notice.

Did You Know?

Some weather data has been recorded in Australia for over 100 years! Scientists use this very long-term data to understand how our climate is changing over time, not just what the weather will do tomorrow.

Find It, Think About It & Apply**Q1 — Find It (Literal)**

- a) What is data? _____
- b) What three things does a meteorologist record daily? _____
- c) What does a wildlife ranger count on the beach? _____

Q2 — Think About It (Inferred)

- a) Why is data collected over many years more useful than data from just one day?

- b) Why might a doctor want data from many patients, not just one?

Q3 — Sort the Job to the Data

Write which job collects each type of data.

Data collected	Job
Temperature, wind speed, rainfall	
Number of turtles hatching each year	
How many patients recover after treatment	

Q4 — Data or Not Data?

Statement	Data / Not Data
"It rained for 3 days this week, according to our rain gauge."	
"I think it might rain because the sky looks a bit grey."	
"We counted 42 turtle hatchlings on the beach last night."	

Watch Out!

A guess or a feeling is not the same as data. Data comes from actually observing, measuring or counting something — it can be checked and verified by others.

Q5 — True or False?

Statement	True / False
Scientists only need data from one day to make a reliable prediction.	True / False
Data helps turn a guess into a trustworthy scientific explanation.	True / False

Apply & Reasoning — Use what you've read to explain your thinking.

- 1 Reasoning:** Why is data more trustworthy than just a guess when explaining something scientific?

- 2 Reasoning:** How could a wildlife ranger use several years of turtle-count data to make a prediction about next year?

- 3 Open-ended:** Choose something you could collect data about at school or at home (e.g. favourite lunch, number of birds seen, minutes of rain). Describe what data you would collect and how you would use it to make a prediction.

Quick Check — Exit Ticket

In one sentence, explain the difference between a guess and data.

Answer & Teacher Support

Answer Key — Pages 1 & 2

Q1 (p2): a) Information gathered from observation, measurement or experiments. b) Temperature, wind speed and rainfall. c) How many turtles hatch.

Q2 (p2): a) Data over many years shows patterns and trends, not just one unusual day. b) Data from many patients shows whether a treatment reliably works, not just for one person by chance.

Q3 (p2): Temperature/wind/rainfall — Meteorologist.
Turtle hatchling count — Wildlife ranger. Patient recovery — Doctor/health scientist.

Q4 (p2): Rain gauge reading — Data. "I think it might rain" — Not data (a guess). Turtle count — Data.

Q5 (p2): "One day's data is enough" — False. "Data turns a guess into a trustworthy explanation" — True.

Answer Key — Page 3

Q1: Data is based on actual observation or measurement and can be checked by others, while a guess is just an unverified idea.

Q2: By comparing the yearly counts, the ranger can see whether numbers are going up, down, or staying steady, and predict whether the trend will likely continue.

Q3 (acceptable-answer guide): Accept any realistic data-collection idea with a clear method and a sensible prediction based on that data.

Exit ticket: A guess is an unverified idea, while data comes from actual observation or measurement and can be checked.

Teacher Checklist

- ☐ Retrieves literal facts from the nonfiction text.
- ☐ Makes reasonable inferences beyond the text.
- ☐ Matches a data type to the job that collects it.
- ☐ Distinguishes data from an unverified guess.
- ☐ Designs a simple, realistic data-collection idea.

Teacher Tip

This resource combines nonfiction reading comprehension with AC9S3H01 (how people use data to develop explanations) — a genuine curriculum gap this fills. Pair with a simple class data-collection activity (e.g. tallying favourite fruit) if time allows.